

21-4c How Do Interest Rates Affect Household Saving?

An important decision that every person faces is how much income to consume today and how much to save for the future. We can use the theory of consumer choice to analyze how people make this decision and how the amount they save depends on the interest rate their savings will earn.

Consider the decision facing Carlos, a worker planning for retirement. To keep things simple, let's divide Carlos's life into two periods. In the first period, Carlos is young and working. In the second period, he is old and retired. When young, Carlos earns **\$100,000**. He divides this income between current consumption and saving. When he is old, Carlos will consume what he has saved, including the interest that his savings have earned.

We can view "consumption when young" and "consumption when old" as the two goods that Carlos must choose between. The interest rate determines the relative price of these two goods. Suppose the interest rate is 10 percent. Then for every dollar that Carlos saves when young, he can consume **\$1.10** when old.

Figure 17 shows Carlos's budget constraint. If he saves nothing, he consumes **\$100,000** when young and nothing when old. If he saves everything, he consumes nothing when young and **\$110,000** when old. The budget constraint shows these and all the intermediate possibilities.

Figure 17

The Consumption-Saving Decision

This figure shows the budget constraint for a person deciding how much to consume in the two periods of his life, the indifference curves representing his preferences, and the optimum.

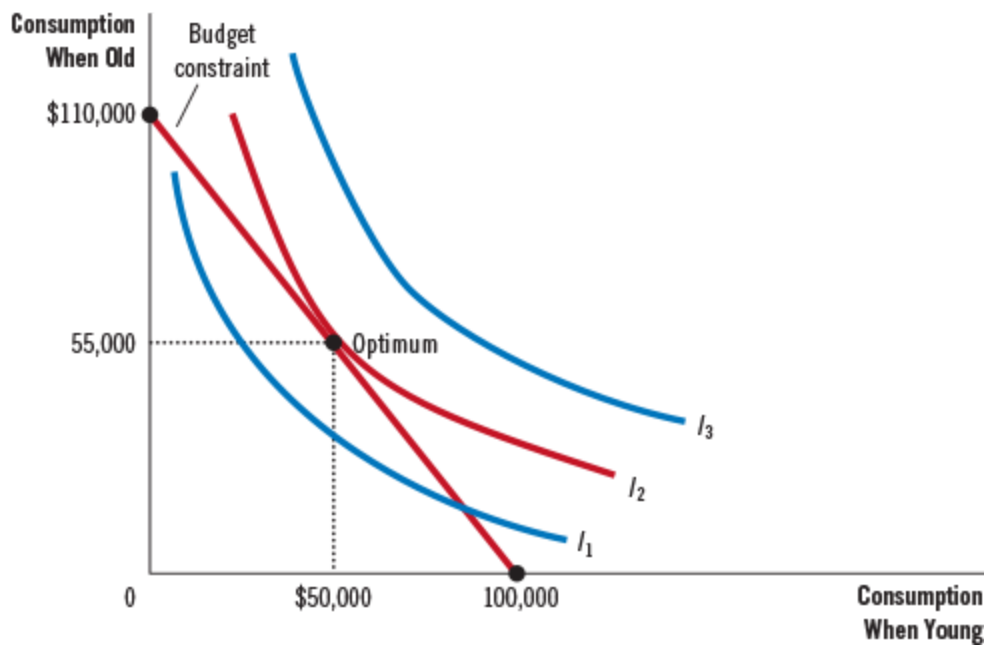


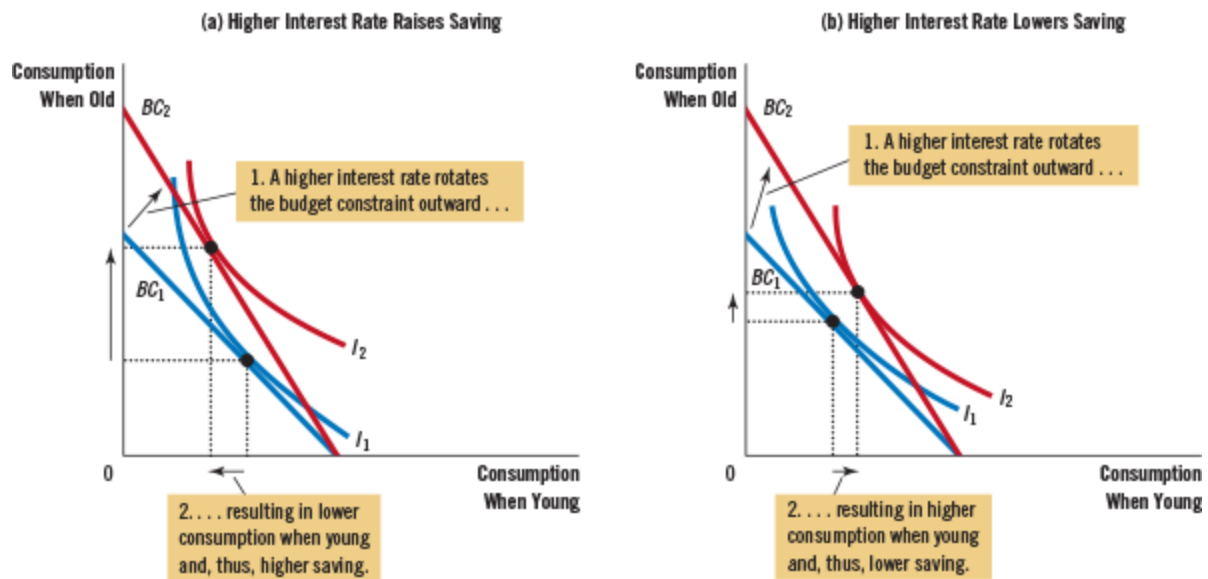
Figure 17 uses indifference curves to represent Carlos's preferences for consumption in the two periods. Because Carlos prefers more consumption in both periods, he prefers points on higher indifference curves to points on lower ones. Given his preferences, Carlos chooses the optimal combination of consumption in both periods of life, which is the point on the budget constraint that is on the highest possible indifference curve. At this optimum, Carlos consumes \$50,000 when young and \$55,000 when old.

Now consider what happens when the interest rate increases from 10 percent to 20 percent. Figure 18 shows two possible outcomes. In both cases, the budget constraint shifts outward and becomes steeper. At the new, higher interest rate, Carlos gets more consumption when old for every dollar of consumption that he gives up when young.

Figure 18

An Increase in the Interest Rate

In both panels, an increase in the interest rate shifts the budget constraint outward. In panel (a), consumption when young falls, and consumption when old rises. The result is an increase in saving when young. In panel (b), consumption in both periods rises. The result is a decrease in saving when young.



The two panels show the results given different preferences by Carlos. In both cases, consumption when old rises. Yet the responses of consumption when young to the change in the interest rate are different in the two cases. In panel (a), Carlos responds to the higher interest rate by consuming less when young. In panel (b), Carlos responds by consuming more when young.

Carlos's saving is his income when young minus his consumption when young. In panel (a), an increase in the interest rate reduces consumption when young, so saving must rise. In panel (b), an increase in the interest rate increases consumption when young, so saving must fall.

The case shown in panel (b) might at first seem odd: Carlos responds to an increase in the return to saving by saving less. But this behavior is not as peculiar as it might seem. We can understand it by considering the income and substitution effects of a higher interest rate.

Consider first the substitution effect. When the interest rate rises, consumption when old becomes less costly relative to consumption when young. Therefore, the substitution effect induces Carlos to consume more when old and less when young. In other words, the substitution effect induces Carlos to save more.

Now consider the income effect. When the interest rate rises, Carlos moves to a higher indifference curve, so he is now better off. As long as consumption when young and consumption when old are both normal goods, he will want to use his increased well-being to enjoy higher consumption in both periods. In other words, the income effect induces him to save less.

The result depends on both the income and substitution effects. If the substitution effect of a higher interest rate is greater than the income effect, Carlos saves more. If the income effect is greater than the substitution effect, Carlos saves less. Thus, the theory of consumer choice says that an increase in the interest rate could either encourage or discourage saving.

This ambiguous result is interesting from the standpoint of economic theory, but it is disappointing from the standpoint of economic policy. It turns out that an important issue in tax policy hinges in part on how saving responds to interest rates. Some economists have advocated reducing the taxation of interest and other capital income, arguing that such a policy change would raise the after-tax interest rate that savers can earn and thereby encourage people to save more. Other economists have argued that because of offsetting income and substitution effects, such a tax change might not increase saving and could even reduce it. Unfortunately, research has not led to a consensus about how interest rates affect saving. As a result, there remains disagreement among economists about whether changes in tax policy aimed at increasing saving would, in fact, have the intended effect.

QuickQuiz

8. Mr. Burns buys only lobster and chicken. Lobster is a normal good, while chicken is an inferior good. When the price of lobster rises, Mr. Burns buys
- a. less of both goods.
 - b. more lobster and less chicken.
 - c. less lobster and more chicken.
 - d. less lobster, but the impact on chicken is ambiguous.
9. If Edna buys more pasta when the price of pasta increases, we can infer that for Edna
- a. pasta is a normal good for which the income effect exceeds the substitution effect.
 - b. pasta is a normal good for which the substitution effect exceeds the income effect.
 - c. pasta is an inferior good for which the income effect exceeds the substitution effect.
 - d. pasta is an inferior good for which the substitution effect exceeds the income effect.
10. Maude's labor-supply curve slopes upward if, for Maude,
- a. leisure is a normal good.

- b. consumption is a normal good.
 - c. the income effect on leisure exceeds the substitution effect.
 - d. the substitution effect on leisure exceeds the income effect.
11. Consumption when young and consumption when old are both normal goods for Seymour, a worker saving for retirement. When the interest rate falls, what happens to Seymour's consumption when old?
- a. It definitely increases.
 - b. It definitely decreases.
 - c. It increases only if the substitution effect exceeds the income effect.
 - d. It decreases only if the substitution effect exceeds the income effect.

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Printed By: Mike Benton (mbento6@wgu.edu)

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